



HK IMO Training 2019-2020
Problem Set 7
For 7 December 2019



Problem 25. There are 315 marbles divided into three piles of 81, 115 and 119. In each move Pythagoras can either merge several piles into a single pile or divide a pile with an even number of marbles into 2 equal piles. Can Pythagoras divide the marbles into 315 piles, each with a single marble?

Problem 26. Each point A in the plane is assigned a real number $f(A)$. It is known that

$$f(M) = f(A) + f(B) + f(C)$$

whenever M is the centroid of $\triangle ABC$. Prove that $f(A) = 0$ for every point A .

Problem 27. Find all positive integers d with the following property: there exists integer $k \geq 3$ such that the numbers $d, 2d, 3d, \dots, kd$ can be ordered in a sequence such that the sum of every two consecutive terms is a perfect square.

Problem 28. Let ABC be an acute-angled triangle. Denote, respectively, by

- k_a, k_b , and k_c the lengths of the symmedians from A, B and C to BC, CA and AB ;
- r_a, r_b , and r_c the exradii of the excircles opposite A, B and C ;
- R and r the circumradius and inradius of $\triangle ABC$.

Prove that

$$k_a(r + r_a) + k_b(r + r_b) + k_c(r + r_c) \leq 4(R + r)^2.$$